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### Loss-controlled computing cable

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#### Abstract

A computing cable that connects computing elements, including: a trace including a first trace segment and a second trace segment, the trace having a first impedance; and an attenuator connecting the first trace segment to the second trace segment, the attenuator including: a resistor having a resistance, and a conductor having a second impedance, wherein the combination of the resistance and the second impedance is based on the first impedance.

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## Background/Summary

### BACKGROUND

#### Field of the Disclosure

(1) The disclosure relates generally to computing cables.

#### Description of the Related Art

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option available to users is information handling systems. An information handling system generally processes, compiles, stores, and/or communicates information or data for business, personal, or other purposes thereby allowing users to take advantage of the value of the information. Because technology and information handling needs and requirements vary between different users or applications, information handling systems may also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information may be processed, stored, or communicated. The variations in information handling systems allow for information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems may include a variety of hardware and software components that may be configured to process, store, and communicate information and may include one or more computer systems, data storage systems, and networking systems.

### SUMMARY

(3) Innovative aspects of the subject matter described in this specification may be embodied in a computing cable that connects computing elements, including: a trace including a first trace segment and a second trace segment, the trace having a first impedance; and an attenuator connecting the first trace segment to the second trace segment, the attenuator including: a resistor having a resistance, and a conductor having a second impedance, wherein the combination of the resistance and the second impedance is based on the first impedance.

(4) Other embodiments of these aspects include corresponding systems and apparatus.

(5) These and other embodiments may each optionally include one or more of the following

features. For instance, the combination of the resistance and the second impedance is equal to the first impedance. The resistor and the conductor are connected in parallel. The conductor is a loop. A length of the loop of the conductor is based on a wavelength of a signal transmitted by the computing cable. The length of the conductor is one-quarter of the wavelength of the signal. The signal is a single-ended signal. The computing cable connects two printed circuit boards within an information handling system. The computing cable connects two information handling systems.

(6) The details of one or more embodiments of the subject matter described in this specification are set forth in the accompanying drawings and the description below. Other potential features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a block diagram of selected elements of an embodiment of an information handling system.
- (2) FIG. 2 illustrates a block diagram of a computing environment, including a computing cable in a first implementation.
- (3) FIG. 3A illustrates a top down perspective view of the computing cable, in the first implementation.
- (4) FIG. 3B illustrates a bottom up perspective view of the computing cable, in the first implementation.
- (5) FIG. 4 illustrates a block diagram of a computing environment, including the computing cable in a second implementation.
- (6) FIG. 5 illustrates a top down perspective view of the computing cable, in the second implementation.
- (7) FIG. 6 illustrates a paddle board layout.

### DESCRIPTION OF PARTICULAR EMBODIMENT(S)

- (8) This disclosure discusses a computing cable. In short, the computing cable can connect together printed circuit boards (PCBs) within an information handling system, or connect together separate information handling systems. The computing cable include resistive loss such that uniform loss can be provided independent of a cable length of the computing cable.
- (9) Specifically, this disclosure discusses a computing cable that connects computing elements, including: a trace including a first trace segment and a second trace segment, the trace having a first impedance; an attenuator connecting the first trace segment to the second trace segment, the attenuator including: a resistor having a resistance, and a conductor having a second impedance, wherein the combination of the resistance and the second impedance is based on the first impedance.
- (10) In the following description, details are set forth by way of example to facilitate discussion of the disclosed subject matter. It should be apparent to a person of ordinary skill in the field, however, that the disclosed embodiments are exemplary and not exhaustive of all possible embodiments.
- (11) For the purposes of this disclosure, an information handling system may include an instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize various forms of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, an information handling system may be a personal computer, a PDA, a consumer electronic device, a network storage device, or another suitable device and may vary in size, shape, performance, functionality, and price. The information

handling system may include memory, one or more processing resources such as a central processing unit (CPU) or hardware or software control logic. Additional components of the information handling system may include one or more storage devices, one or more communications ports for communicating with external devices as well as various input and output (I/O) devices, such as a keyboard, a mouse, and a video display. The information handling system may also include one or more buses operable to transmit communication between the various hardware components.

(12) For the purposes of this disclosure, computer-readable media may include an instrumentality or aggregation of instrumentalities that may retain data and/or instructions for a period of time. Computer-readable media may include, without limitation, storage media such as a direct access storage device (e.g., a hard disk drive or floppy disk), a sequential access storage device (e.g., a tape disk drive), compact disk, CD-ROM, DVD, random access memory (RAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), and/or flash memory (SSD); as well as communications media such wires, optical fibers, microwaves, radio waves, and other electromagnetic and/or optical carriers; and/or any combination of the foregoing.

(13) Particular embodiments are best understood by reference to FIGS. **1-6** wherein like numbers are used to indicate like and corresponding parts.

(14) Turning now to the drawings, FIG. **1** illustrates a block diagram depicting selected elements of an information handling system **100** in accordance with some embodiments of the present disclosure. In various embodiments, information handling system **100** may represent different types of portable information handling systems, such as, display devices, head mounted displays, head mount display systems, smart phones, tablet computers, notebook computers, media players, digital cameras, 2-in-1 tablet-laptop combination computers, and wireless organizers, or other types of portable information handling systems. In one or more embodiments, information handling system **100** may also represent other types of information handling systems, including desktop computers, server systems, controllers, and microcontroller units, among other types of information handling systems. Components of information handling system **100** may include, but are not limited to, a processor subsystem **120**, which may comprise one or more processors, and system bus **121** that communicatively couples various system components to processor subsystem **120** including, for example, a memory subsystem **130**, an I/O subsystem **140**, a local storage resource **150**, and a network interface **160**. System bus **121** may represent a variety of suitable types of bus structures, e.g., a memory bus, a peripheral bus, or a local bus using various bus architectures in selected embodiments. For example, such architectures may include, but are not limited to, Micro Channel Architecture (MCA) bus, Industry Standard Architecture (ISA) bus, Enhanced ISA (EISA) bus, Peripheral Component Interconnect (PCI) bus, PCI-Express bus, HyperTransport (HT) bus, and Video Electronics Standards Association (VESA) local bus.

(15) As depicted in FIG. **1**, processor subsystem **120** may comprise a system, device, or apparatus operable to interpret and/or execute program instructions and/or process data, and may include a microprocessor, microcontroller, digital signal processor (DSP), application specific integrated circuit (ASIC), or another digital or analog circuitry configured to interpret and/or execute program instructions and/or process data. In some embodiments, processor subsystem **120** may interpret and/or execute program instructions and/or process data stored locally (e.g., in memory subsystem **130** and/or another component of information handling system). In the same or alternative embodiments, processor subsystem **120** may interpret and/or execute program instructions and/or process data stored remotely (e.g., in network storage resource **170**).

(16) Also in FIG. **1**, memory subsystem **130** may comprise a system, device, or apparatus operable to retain and/or retrieve program instructions and/or data for a period of time (e.g., computer-readable media). Memory subsystem **130** may comprise random access memory (RAM), electrically erasable programmable read-only memory (EEPROM), a PCMCIA card, flash memory, magnetic storage, opto-magnetic storage, and/or a suitable selection and/or array of volatile or non-

volatile memory that retains data after power to its associated information handling system, such as system **100**, is powered down.

(17) In information handling system **100**, I/O subsystem **140** may comprise a system, device, or apparatus generally operable to receive and/or transmit data to/from/within information handling system **100**. I/O subsystem **140** may represent, for example, a variety of communication interfaces, graphics interfaces, video interfaces, user input interfaces, and/or peripheral interfaces. In various embodiments, I/O subsystem **140** may be used to support various peripheral devices, such as a touch panel, a display adapter, a keyboard, an accelerometer, a touch pad, a gyroscope, an IR sensor, a microphone, a sensor, or a camera, or another type of peripheral device.

(18) Local storage resource **150** may comprise computer-readable media (e.g., hard disk drive, floppy disk drive, CD-ROM, and/or other type of rotating storage media, flash memory, EEPROM, and/or another type of solid state storage media) and may be generally operable to store instructions and/or data. Likewise, the network storage resource may comprise computer-readable media (e.g., hard disk drive, floppy disk drive, CD-ROM, and/or other type of rotating storage media, flash memory, EEPROM, and/or other type of solid state storage media) and may be generally operable to store instructions and/or data.

(19) In FIG. **1**, network interface **160** may be a suitable system, apparatus, or device operable to serve as an interface between information handling system **100** and a network **110**. Network interface **160** may enable information handling system **100** to communicate over network **110** using a suitable transmission protocol and/or standard, including, but not limited to, transmission protocols and/or standards enumerated below with respect to the discussion of network **110**. In some embodiments, network interface **160** may be communicatively coupled via network **110** to a network storage resource **170**. Network **110** may be a public network or a private (e.g. corporate) network. The network may be implemented as, or may be a part of, a storage area network (SAN), personal area network (PAN), local area network (LAN), a metropolitan area network (MAN), a wide area network (WAN), a wireless local area network (WLAN), a virtual private network (VPN), an intranet, the Internet or another appropriate architecture or system that facilitates the communication of signals, data and/or messages (generally referred to as data). Network interface **160** may enable wired and/or wireless communications (e.g., NFC or Bluetooth) to and/or from information handling system **100**.

(20) In particular embodiments, network **110** may include one or more routers for routing data between client information handling systems **100** and server information handling systems **100**. A device (e.g., a client information handling system **100** or a server information handling system **100**) on network **110** may be addressed by a corresponding network address including, for example, an Internet protocol (IP) address, an Internet name, a Windows Internet name service (WINS) name, a domain name or other system name. In particular embodiments, network **110** may include one or more logical groupings of network devices such as, for example, one or more sites (e.g. customer sites) or subnets. As an example, a corporate network may include potentially thousands of offices or branches, each with its own subnet (or multiple subnets) having many devices. One or more client information handling systems **100** may communicate with one or more server information handling systems **100** via any suitable connection including, for example, a modem connection, a LAN connection including the Ethernet or a broadband WAN connection including DSL, Cable, T1, T3, Fiber Optics, Wi-Fi, or a mobile network connection including GSM, GPRS, 3G, or WiMax.

(21) Network **110** may transmit data using a desired storage and/or communication protocol, including, but not limited to, Fibre Channel, Frame Relay, Asynchronous Transfer Mode (ATM), Internet protocol (IP), other packet-based protocol, small computer system interface (SCSI), Internet SCSI (iSCSI), Serial Attached SCSI (SAS) or another transport that operates with the SCSI protocol, advanced technology attachment (ATA), serial ATA (SATA), advanced technology attachment packet interface (ATAPI), serial storage architecture (SSA), integrated drive electronics (IDE), and/or any combination thereof. Network **110** and its various components may be

implemented using hardware, software, or any combination thereof.

(22) Turning to FIG. 2, FIG. 2 illustrates an environment **200** including a computing cable **210** (or wire **210**), a first computing element **212a**, and a second computing element **212b**. The first computing element **212a** and the second computing element **212b** can be referred to as computing elements **212**. The computing cable **210** can include a trace **220** and an attenuator **222**; however, the computing cable **210** can include any number of traces and attenuators.

(23) In some examples, the computing elements **212** are information handling systems, and can be similar to, or includes, the information handling system **100** of FIG. 1. In some examples, the computing elements **212** are printed circuit boards (PCBs) of an information handling system (e.g., the information handling system **100**).

(24) The computing cable **210** can connect the first computing element **212a** to the second computing element **212b**. When the computing elements **212** include information handling systems, the computing cable **210** connects the two information handling systems. When the computing elements **212** include PCBs, the computing cable **210** connects the PCBs.

(25) FIG. 3A illustrates a top down perspective view of the computing cable **210**; and FIG. 3B illustrates a bottom up perspective view of the computing cable **210**. The cable **210** (or wire **210**) can include the trace **220**. The trace **220** can include a first trace segment **302a** and a second trace segment **302b** (collectively referred to as trace segments **302**). The trace **220** can be associated with a first impedance  $Z_{\text{sub.1}}$ . That is, the first trace segment **302a** can be associated with the first impedance  $Z_{\text{sub.1}}$ ; and the second trace segment **302b** can be associated with the first impedance  $Z_{\text{sub.1}}$ .

(26) The attenuator **222** connects the first trace segment **302a** to the second trace segment **302b**. The attenuator **222** includes a resistor **304** and a conductor **306**. The resistor **304** can be associated with a first resistance  $R_{\text{sub.1}}$ , and the conductor **306** can be associated with a second impedance  $Z_{\text{sub.2}}$ . In some examples, the value of the first resistance  $R_{\text{sub.1}}$  is chosen based on a desired loss of the computing cable **210**. A lower value of the first resistance  $R_{\text{sub.1}}$  results in increase in the loss of the computing cable **210**.

(27) The resistor **304** and the conductor **306** are connected in parallel along the trace **220**. Furthermore, the conductor **306** is a loop having a length. For example, the length of the loop of the conductor **306** is defined as the total distance added by the loop between the first trace segment **302a** and the second trace segment **302b**. For example, the length of the loop of the conductor **306** can be defined between the points **320a** and **320b** around the loop. For example, the length of the loop of the conductor **306** can be defined by the distance **322**.

(28) In some examples, the length of the loop of the conductor **306** is based on a wavelength of a signal transmitted by the computing cable **210**. That is, the length of the loop of the conductor **306** is based on a wavelength of a signal transmitted along the signal trace segments **302**. In some examples, the length of the loop of the conductor **306** is one-quarter of the wavelength of the signal. For example, when the wavelength of the signal is 16 GHz, the length of the loop of the conductor **306** is 3 millimeters.

(29) To that end, the combination of the first resistance  $R_{\text{sub.1}}$  and the second impedance  $Z_{\text{sub.2}}$  is based on the first impedance  $Z_{\text{sub.1}}$ . That is, the combination of the first resistance  $R_{\text{sub.1}}$  of the resistor **304** and the second impedance  $Z_{\text{sub.2}}$  of the conductor **306** is based on the first impedance  $Z_{\text{sub.1}}$  of the first trace segment **302a** or the second trace segment **302b**. That is, the first resistance  $R_{\text{sub.1}}$  of the resistor **304** and the second impedance  $Z_{\text{sub.2}}$  of the conductor **306** should “match” the first impedance  $Z_{\text{sub.1}}$  of the first trace segment **302a** or the second trace segment **302b**.

(30) In some examples, the first resistance  $R_{\text{sub.1}}$  and the second impedance  $Z_{\text{sub.2}}$  equal the first impedance  $Z_{\text{sub.1}}$ . That is, as the resistor **304** and the conductor **306** are connected in parallel, the length of the loop of the conductor **306** and the values of the first resistance  $R_{\text{sub.1}}$  of the first resistor **304** and the second impedance  $Z_{\text{sub.2}}$  of the conductor **306** are chosen such that

Z.sub.1=Rill Z.sub.2.

(31) In some examples, the signal that is transmitted by the computing cable **210** is a single-ended signal. For example, the signal that is transmitted by the computing cable **210** is a DDR signal.

(32) Turning to FIG. 4, FIG. 4 illustrates an environment **400** including a computing cable **410** (or wire **410**), a first computing element **412a**, and a second computing element **412b**. The first computing element **412a** and the second computing element **412b** can be referred to as computing elements **412**.

(33) The computing cable **410** can include a first signal transmission element **411a** and a second signal transmission element **411b**. The first signal transmission element **411a** and the second signal transmission element **411b** can collectively be referred to as signal transmission elements **411**.

(34) The first signal transmission element **411a** can include a first trace **420a** and a first attenuator **422a**; however, the computing cable **410** can include any number of traces and attenuators. The second signal transmission element **411b** can include a second trace **420b** and a second attenuator **422b**; however, the computing cable **410** can include any number of traces and attenuators.

(35) In some examples, the computing elements **412** are information handling systems, and can be similar to, or includes, the information handling system **100** of FIG. 1. In some examples, the computing elements **412** are a printed circuit boards (PCBs) of an information handling system (e.g., the information handling system **100**).

(36) The computing cable **410** can connect the first computing element **412a** to the second computing element **412b**. When the computing elements **412** include information handling systems, the computing cable **410** connects the two information handling systems. When the computing elements **412** include PCBs, the computing cable **410** connects the PCBs.

(37) FIG. 5 illustrates a top down perspective view of the computing cable **410**. The cable **410** (or wire **410**) can include the first signal transmission element **411a** and the second signal transmission element **411b**.

(38) With respect to the first signal transmission element **411a**, the first trace **420a** of the first signal transmission element **411a** can include a first trace segment **502a** and a second trace segment **502b**. The first trace **420a** can be associated with the first impedance Z.sub.1. That is, the first trace segment **502a** can be associated with the first impedance Z.sub.1; and the second trace segment **502b** can be associated with the first impedance Z.sub.1.

(39) The first attenuator **422a** connects the first trace segment **502a** to the second trace segment **502b**. The first attenuator **422a** includes a first resistor **504a** and a first conductor **506a**. The first resistor **504a** can be associated with the first resistance R.sub.1, and the first conductor **506a** can be associated with the second impedance Z.sub.2. In some examples, the value of the first resistance R.sub.1 is chosen based on a desired loss of the computing cable **410**. A lower value of the first resistance R.sub.1 results in increase in the loss of the computing cable **410**.

(40) The first resistor **504a** and the first conductor **506a** are connected in parallel along the trace **420a**. Furthermore, the first conductor **506a** is a loop having a length. For example, the length of the loop of the first conductor **506a** is defined as the total distance added by the loop between the first trace segment **502a** and the second trace segment **502b**. For example, the length of the loop of the first conductor **506a** can be defined similar to that with respect to the conductor **306** (e.g., between the points **320a** and **320b** around the loop; by the distance **322**).

(41) In some examples, the length of the loop of the first conductor **506a** is based on a wavelength of a signal transmitted by the computing cable **410**. That is, the length of the loop of the first conductor **506a** is based on a wavelength of a signal transmitted along the signal trace segments **502a** and **502b**. In some examples, the length of the loop of the first conductor **506a** is one-quarter of the wavelength of the signal. For example, when the wavelength of the signal is 16 GHz, the length of the loop of the first conductor **506a** is 3 millimeters.

(42) To that end, the combination of the first resistance R.sub.1 and the second impedance Z.sub.2 is based on the first impedance Z.sub.1. That is, the combination of the first resistance R.sub.1 of

the first resistor **504a** and the second impedance  $Z_{\text{sub.2}}$  of the first conductor **506a** is based on the first impedance  $Z_{\text{sub.1}}$  of the first trace segment **502a** or the second trace segment **502b**. That is, the first resistance  $R_{\text{sub.1}}$  of the first resistor **504a** and the second impedance  $Z_{\text{sub.2}}$  of the first conductor **506a** should “match” the first impedance  $Z_{\text{sub.1}}$  of the first trace segment **502a** or the second trace segment **502b**.

(43) In some examples, the first resistance  $R_{\text{sub.1}}$  and the second impedance  $Z_{\text{sub.2}}$  equals the first impedance  $Z_{\text{sub.1}}$ . That is, as the first resistor **504a** and the first conductor **506a** are connected in parallel, the length of the loop of the first conductor **506a** and the values of the first resistance  $R_{\text{sub.1}}$  of the first resistor **504a** and the second impedance  $Z_{\text{sub.2}}$  of the first conductor **506a** are chosen such that  $Z_{\text{sub.1}} = R_{\text{sub.1}} \parallel Z_{\text{sub.2}}$ .

(44) With respect to the second signal transmission element **411b**, the second trace **420b** of the second signal transmission element **411b** can include a third trace segment **502c** and a fourth trace segment **502d**. The second trace **420b** can be associated with a third impedance  $Z_{\text{sub.3}}$ . That is, the third trace segment **502c** can be associated with the third impedance  $Z_{\text{sub.3}}$ ; and the fourth trace segment **502d** can be associated with the third impedance  $Z_{\text{sub.3}}$ .

(45) The second attenuator **422b** connects the third trace segment **502c** to the fourth trace segment **502d**. The second attenuator **422b** includes a second resistor **504b** and a second conductor **506b**. The second resistor **504b** can be associated with a second resistance  $R_{\text{sub.2}}$ , and the second conductor **506b** can be associated with a fourth impedance  $Z_{\text{sub.4}}$ . In some examples, the value of the second resistance  $R_{\text{sub.2}}$  is chosen based on a desired loss of the computing cable **410**. A lower value of the second resistance  $R_{\text{sub.2}}$  results in increase in the loss of the computing cable **410**.

(46) The second resistor **504b** and the second conductor **506b** are connected in parallel along the trace **420b**. Furthermore, the second conductor **506b** is a loop having a length. For example, the length of the loop of the second conductor **506b** is defined as the total distance added by the loop between the third trace segment **502c** and the fourth trace segment **502d**. For example, the length of the loop of the second conductor **506b** can be defined similar to that with respect to the conductor **306** (e.g., between the points **320a** and **320b** around the loop; by the distance **322**).

(47) In some examples, the length of the loop of the second conductor **506b** is based on a wavelength of a signal transmitted by the computing cable **410**. That is, the length of the loop of the second conductor **506b** is based on a wavelength of a signal transmitted along the signal trace segments **502c** and **502d**. In some examples, the length of the loop of the second conductor **506b** is one-quarter of the wavelength of the signal. For example, when the wavelength of the signal is 16 GHz, the length of the loop of the second conductor **506b** is 3 millimeters.

(48) To that end, the combination of the second resistance  $R_{\text{sub.2}}$  and the fourth impedance  $Z_{\text{sub.4}}$  is based on the third impedance  $Z_{\text{sub.3}}$ . That is, the combination of the second resistance  $R_{\text{sub.2}}$  of the second resistor **504b** and the fourth impedance  $Z_{\text{sub.4}}$  of the second conductor **506b** is based on the third impedance  $Z_{\text{sub.3}}$  of the third trace segment **502c** or the fourth trace segment **502d**. That is, the second resistance  $R_{\text{sub.2}}$  of the second resistor **504b** and the fourth impedance  $Z_{\text{sub.4}}$  of the second conductor **506b** should “match” the third impedance  $Z_{\text{sub.3}}$  of the third trace segment **502c** or the fourth trace segment **502d**.

(49) In some examples, the second resistance  $R_{\text{sub.2}}$  and the fourth impedance  $Z_{\text{sub.4}}$  equals the third impedance  $Z_{\text{sub.3}}$ . That is, as the second resistor **504b** and the second conductor **506b** are connected in parallel, the length of the loop of the second conductor **506b** and the values of the second resistance  $R_{\text{sub.2}}$  of the second resistor **504b** and the fourth impedance  $Z_{\text{sub.4}}$  of the second conductor **506b** are chosen such that  $Z_{\text{sub.3}} = R_{\text{sub.2}} \parallel Z_{\text{sub.4}}$ .

(50) In some examples, the signal that is transmitted by the computing cable **410** is a differential signal. For example, the signal that is transmitted by the computing cable **410** is a PCIe signal.

(51) FIG. 6 illustrates a paddle board layout **602** (high-level layout of a paddle board-based implementation). For example, the paddle board **602** can include multiple computing cables **610**,



which can be the computing cable **210** and/or the computing cable **410**. The paddle board **602** can include a ribbon cable **620**.

(52) The above disclosed subject matter is to be considered illustrative, and not restrictive, and the appended claims are intended to cover all such modifications, enhancements, and other embodiments which fall within the true spirit and scope of the present disclosure. Thus, to the maximum extent allowed by law, the scope of the present disclosure is to be determined by the broadest permissible interpretation of the following claims and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

(53) Herein, “or” is inclusive and not exclusive, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A or B” means “A, B, or both,” unless expressly indicated otherwise or indicated otherwise by context. Moreover, “and” is both joint and several, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A and B” means “A and B, jointly or severally,” unless expressly indicated otherwise or indicated otherwise by context.

(54) The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes and illustrates respective embodiments herein as including particular components, elements, features, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, features, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend. Furthermore, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative.

## Claims

1. A computing cable that connects computing elements, including: a trace including a first trace segment and a second trace segment, the trace having a first impedance; and an attenuator connecting the first trace segment to the second trace segment, the attenuator including: a resistor having a resistance, and a conductor having a second impedance, wherein the combination of the resistance and the second impedance is equal to the first impedance.
2. The computing cable of claim 1, wherein the resistor and the conductor are connected in parallel.
3. The computing cable of claim 1, wherein the conductor is a loop.
4. The computing cable of claim 3, wherein a length of the loop of the conductor is based on a wavelength of a signal transmitted by the computing cable.
5. The computing cable of claim 4, wherein the length of the conductor is one-quarter of the wavelength of the signal.
6. The computing cable of claim 4, wherein the signal is a single-ended signal.
7. The computing cable of claim 1, wherein the computing cable connects two printed circuit boards within an information handling system.
8. The computing cable of claim 1, wherein the computing cable connects two information handling systems.
9. A computing cable that connects computing elements, including: a first signal transmission element, including: a first trace including a first trace segment and a second trace segment, the first trace having a first impedance; and a first attenuator connecting the first trace segment to the

second trace segment, the first attenuator including: a first resistor having a first resistance, and a first conductor having a second impedance, a second signal transmission element, including: a second trace including a third trace segment and a fourth trace segment, the second trace having a third impedance; and a second attenuator connecting the third trace segment to the fourth trace segment, the second attenuator including: a second resistor having a second resistance, and a second conductor having a fourth impedance, wherein the combination of the first resistance and the second impedance is equal to the first impedance, wherein the combination of the second resistance and the fourth impedance is equal to the third impedance.

10. The computing cable of claim 9, wherein the first resistor and the first conductor are connected in parallel, and wherein the second resistor and the second conductor are connected in parallel.

11. The computing cable of claim 9, wherein the first conductor is a first loop, and the second conductor is a second loop.

12. The computing cable of claim 11, wherein the length of the first loop of the first conductor and the second loop of the second conductor is based on a wavelength of a signal transmitted by the computing cable.

13. The computing cable of claim 12, wherein the length of the first loop of the first conductor and the second loop of the second conductor is one-quarter of the wavelength of the signal.

14. The computing cable of claim 12, wherein the signal is a differential signal.

15. The computing cable of claim 9, wherein the computing cable connects two printed circuit boards within an information handling system.

16. The computing cable of claim 9, wherein the computing cable connects two information handling systems.

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